

Lecture 11 2026-10-06

Today: terminology of trajectory optimization

Continuous time trajectory optimization problem:

$$\min_{t_0, t_F, x(t), u(t)} \quad J(t_0, t_F, x(t_0), x(t_F); p(t_F)) \\ + \int_{t_0}^{t_F} w(x(\tau), u(\tau); p(\tau)) d\tau$$

terminal cost: $J(t_0, t_F, x(t_0), x(t_F); p(t_F))$

in practice: almost never see t_0 or $x(t_0)$

Lagrange term (stage-wise cost):

$$\int_{t_0}^{t_F} w(x(\tau), u(\tau); p(\tau)) d\tau$$

Decision variables: what you actually solve for

State $x(t) \in \mathbb{R}^{n_x}$

Control $u(t) \in \mathbb{R}^{n_u}$

Times $t_0 \in \mathbb{R}_{++}$ $t_F \in \mathbb{R}_{++}$

final time for trajectory can
be a decision variable

Parameters $p(t) \in \mathbb{R}^{n_p}$

e.g., mass of a spacecraft

e.g., wind speeds that day

$\Rightarrow$ Putting this altogether: let's define the problem $\mathcal{P}$

$$\begin{aligned} \mathcal{P}[p(t)] \quad \min_{t_0, t_F, x(t), u(t)} & \quad J(t_0, t_F, x(t_0), x(t_F); p(t_F)) \\ & + \int_{t_0}^{t_F} w(\tau, x(\tau), u(\tau); p(\tau)) d\tau \end{aligned}$$

subj. to:

$$\dot{x}(t) = f(x(t), u(t); p(t)) \quad \forall t \in [t_0, t_F] \quad \text{system dynamics}$$

$$f(t, x(t), u(t); p(t)) \quad \forall t \in [t_0, t_F] \quad \text{system dynamics}$$

$$g(t, x(t), u(t); p(t)) \leq 0 \quad \forall t \in [t_0, t_F] \quad \text{path constraints}$$

$$g(t_0, t_F, x(t_0), x(t_F); p(t_F)) \leq 0 \quad \text{boundary conditions}$$

Common examples:

Stage costs

$$1. \int_{T_0}^{T_F} \|u(t)\|_{\ell_1} dt$$

Minimum fuel

$$2. \int_{T_0}^{T_F} \|u(t)\|_{\ell_2} dt$$

Minimum energy

$$3. \int_{T_0}^{T_F} 1 dt$$

Minimum time

$$4. \int_{T_0}^{T_F} (x(t) - x_g)^T Q (x(t) - x_g) + u(t)^T R u(t) dt$$

$Q, R \succ 0$

Linear quadratic regulator cost (assumes $\dot{x} = Ax + Bu$)

Path constraints: $g(t, x(t), u(t)) \leq 0$

$$1. x_{\min} \leq x(t) \leq x_{\max}$$

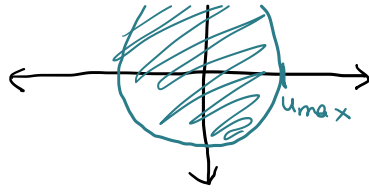
Box constraints

$$2. u_{\min} \leq u(t) \leq u_{\max}$$

$$3. \|u(t)\|_{\ell_2} \leq u_{\max}$$

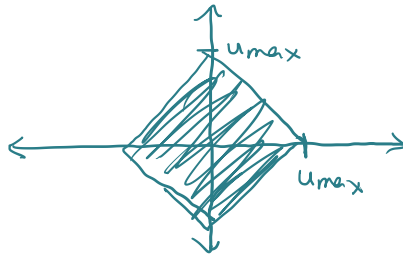
L_2 -norm control constraint





$$4. \|u(t)\|_{\ell_1} \leq u_{\max}$$

L_1 -norm control constraint

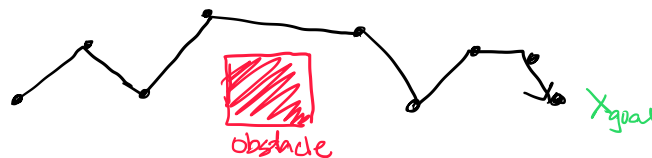


Common classes of traj. opt. problems:

1. Two-point boundary value problem (2PBVP)

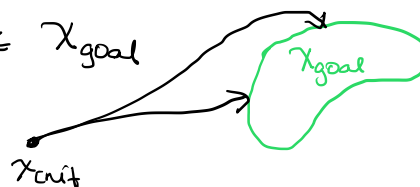
$$x(t_0) = x_{\text{init}} \quad x(t_F) = x_{\text{goal}}$$

→ Used commonly in sampling-based motion planning (SBMP)
to connect two randomly drawn samples



2. Free-final state problem

$$x(t_F) \in X_{\text{goal}}$$



A common free-final state problem: land anywhere

A common free-final state problem: land anywhere within some l_2 -distance of a point x_{goal} :

$$\|x(t_F) - x_{\text{goal}}\|_{l_2} \leq r_{\text{dist}}$$

3. Free-final time problem: t_F is a decision variable

How to convert these continuous time optimization problems into something with a finite number of decision variables?

1. "Optimize and then discretize" $\rightarrow$ uses calculus of variations to solve $\mathcal{P}$ in continuous time
2. "Discretize and then optimize" $\rightarrow$ allows us to use "off-the-shelf" solver after converting $x(t)/u(t)$ to a finite set of variables

$\Rightarrow$ For approach #2: start with $\mathcal{P}$ from above

1. Reformulate $\mathcal{P}$ into a more tractable infinite dimensional form
2. Discretize

e.g., $\dot{x}(t) = f(x(t), u(t)) \rightarrow x_{k+1} = \bar{f}(x_k, u_{k+1})$

e.g., $x_{\min} \leq x(t) \leq x_{\max} \rightarrow x_{\min} \leq x_k \leq x_{\max}$

Pitfall: discretizing a constraint might lead to constraint violation between x_k and x_{k+1}



Doing all of this yields an optimization problem with a finite number of decision variables:

$$\overline{\mathcal{P}}[p_{0:N}] \quad \text{minimize}_{x_{0:N}, u_{0:N}, \Delta t_{0:N}} \underbrace{l_N(x_N)}_{\text{terminal cost}} + \underbrace{\sum_{k=0}^{N-1} l_k(x_k, u_k, \Delta t_k; p_k)}_{\text{stage cost}}$$

subj. to:

$$x_{k+1} = \overline{f}(x_k, u_k, \Delta t_k; p_k) \quad k=0, \dots, N-1$$

$\uparrow$ discretized version of continuous time dynamics $f(x(t), u(t), t)$

$$g_i(x_k, u_k, \Delta t_k; p_k) \leq 0 \quad k=0, \dots, N-1$$

$$i=1, \dots, p$$

$$h_i(x_k, u_k, \Delta t_k; p_k) = 0 \quad k=0, \dots, N-1$$

$$i=1, \dots, m$$

$\Rightarrow$ Now $\overline{\mathcal{P}}$ can be solved using some optimization solver

